

A REPORT ON
PROJECT MANUFACTURING PROJECT
(EGC-381)



DAYALBAGH EDUCATIONAL INSTITUTE
(DEEMED TO BE UNIVERSITY) DAYALBAGH , AGRA

PROJECT TITLE:AUTONOMOUS VACUUM AND MOPPING ROBOT

**SUBMITTED TO :
DR. V PREM PRAKASH**

**SUBMITTED BY :
KHYATI PARASHAR (2400886)
VARALIKA SHARMA (2400993)
B.TECH 2nd YEAR
(ELECTRICAL)**

Contents

Contents	2
ABSTRACT	2
INTRODUCTION	3
PROBLEM STATEMENT	4
OBJECTIVES	5
SYSTEM OVERVIEW	6
HARDWARE COMPONENTS	7
DESIGN AND DEVELOPMENT	9
WORKING PRINCIPLE	11
CIRCUIT & CONNECTION DETAILS.....	11
COMPARISON WITH EXISTING CLEANING ROBOTS	12
APPLICATIONS	13
ADVANTAGES	14
LIMITATIONS.....	15
FUTURE SCOPE	16
CONCLUSION.....	17
REFERENCES	18

ABSTRACT

Maintaining floor cleanliness is essential for hygiene and health, yet manual cleaning remains time-consuming, physically demanding, and impractical for large areas or hazardous environments. Existing commercial robotic cleaners, while effective, are prohibitively expensive (₹20,000-₹38,000) and difficult to repair, limiting their accessibility to average households and small institutions.

This project presents the design and development of a low-cost autonomous vacuum and mopping robot for indoor floor cleaning. The system integrates an Arduino Uno microcontroller, dual HC-SR04 ultrasonic sensors for obstacle detection and avoidance, RFID-based access control, L298N motor driver with four DC motors for mobility, dual 12V vacuum fans for dust collection, and a remote-controlled water pump with wiper mechanism for wet mopping.

The robot operates autonomously, navigating around obstacles while simultaneously performing vacuum and mopping operations. The RFID authentication system prevents unauthorized use, while the remote-controlled water pump allows selective mopping control, preventing water damage to sensitive surfaces and conserving water. The complete system is powered by a 12V rechargeable battery.

The developed prototype demonstrates effective cleaning performance on indoor surfaces at an estimated cost of ₹4,000—approximately 80-90% cheaper than commercial alternatives. The design utilizes readily available components, ensuring easy maintenance and repair without specialized technical expertise. This work validates the feasibility of affordable robotic cleaning automation for households, educational institutions, and small commercial spaces. Future enhancements may include IoT connectivity, advanced navigation algorithms, and self-charging capabilities.

INTRODUCTION

Cleaning is essential in every environment—whether residential homes, industrial facilities, educational institutions, or commercial spaces. In today’s fast-paced world, maintaining cleanliness has become increasingly challenging due to hectic work schedules and time constraints. Regular floor cleaning, though necessary for health and hygiene, often gets neglected due to lack of time or physical capability.

The challenge becomes even more significant in certain scenarios. In large commercial spaces, multiple workers are required for thorough cleaning, increasing labor costs. In hazardous environments such as nuclear facilities, chemical plants, or hospitals dealing with infectious diseases, human presence during cleaning operations poses serious health risks. Workers may be exposed to harmful radiation, toxic chemicals, or biological hazards that can cause severe illness or even death.

Furthermore, human limitations such as fatigue, inconsistent cleaning quality, and the physical strain of repetitive tasks make manual cleaning inefficient for large areas. Elderly individuals and people with physical disabilities find floor cleaning particularly difficult and exhausting.

Automation has emerged as a transformative solution in modern household and industrial applications. Robotic cleaning systems represent a significant advancement in reducing manual labor while improving cleaning efficiency and consistency. These systems can operate continuously without fatigue, maintain consistent cleaning quality, and access areas that might be difficult or dangerous for humans.

This project addresses these challenges by developing an affordable, autonomous robotic cleaning system. The robot combines vacuum cleaning and mopping functionalities, providing comprehensive floor care. Using Arduino Uno as its brain, ultrasonic sensors for navigation, and RFID technology for security, the system demonstrates that advanced cleaning automation doesn’t need to be expensive or complex. The design prioritizes accessibility, allowing users to build, maintain, and repair the system without specialized technical knowledge or expensive replacement parts.

PROBLEM STATEMENT

The modern cleaning challenge is multifaceted, affecting households, institutions, and industries alike. Traditional manual floor cleaning presents several critical problems:

Time and Effort Constraints: Manual cleaning of floors requires significant time and physical effort. In today's busy lifestyle, people struggle to find time for regular, thorough floor cleaning. An average household spends 2-3 hours per week on floor cleaning alone, time that could be utilized more productively.

Health and Safety Concerns: Prolonged bending, kneeling, and repetitive motions during manual cleaning can cause musculoskeletal problems, particularly back pain and knee issues. For elderly individuals or people with physical disabilities, floor cleaning can be nearly impossible without assistance.

Cost of Commercial Solutions: While commercial robotic cleaners exist in the market, they are prohibitively expensive, typically costing ₹20,000 to ₹38,000 or more. This price range makes them inaccessible to average households and small institutions. Additionally, these commercial units often have proprietary parts, making repairs expensive and sometimes requiring complete replacement.

Limited Functionality: Many affordable cleaning devices perform only one function—either vacuuming or mopping—requiring users to purchase multiple devices or perform tasks separately. This increases both cost and cleaning time.

Security and Unauthorized Use: In institutional or commercial settings, cleaning equipment may be used without authorization, leading to misuse, damage, or theft. There's a need for controlled access systems that are simple yet effective.

Water Management Issues: Existing automatic mopping systems often lack user control over water dispensing, leading to over-wetting of floors, water wastage, or damage to water-sensitive surfaces like carpets or certain types of wooden flooring.

Therefore, there exists a clear need for an economical, customizable cleaning robot that addresses all these challenges while remaining affordable, repairable, and effective.

OBJECTIVES

The primary goal of this project is to develop a multifunctional, autonomous cleaning robot that addresses the limitations of both manual cleaning and expensive commercial alternatives. The specific objectives are:

1. **Dual-Function Cleaning System:** To design and implement a robot capable of performing both vacuum cleaning and wet mopping simultaneously. This dual functionality ensures comprehensive floor cleaning in a single pass, saving time and energy.
2. **Intelligent Obstacle Detection and Avoidance:** To implement an effective navigation system using ultrasonic sensors that enables the robot to detect obstacles in real-time and autonomously change direction to avoid collisions. The system should maintain continuous cleaning operation while safely navigating around furniture, walls, and other obstacles.
3. **Secure Access Control:** To provide an RFID-based authentication system that ensures only authorized users can operate the robot. This prevents unauthorized use, potential misuse, and protects the equipment from damage due to improper handling.
4. **User-Controlled Water Management:** To enable selective, manual control of the water pump through a wireless remote control system. This allows users to decide when and where mopping is required, preventing water damage to sensitive areas like carpets, rugs, or electronic equipment, while also conserving water.
5. **Cost-Effective Design:** To develop a robot using readily available, affordable components that keep the total cost under ₹5,000, making it accessible to average households, educational institutions, and small businesses.
6. **Ease of Maintenance and Repair:** To ensure the design uses standard, replaceable components that are easily available in the market. The robot should be simple enough for users to perform basic maintenance and repairs without requiring specialized technical knowledge or expensive service centers.
7. **Energy Efficiency:** To optimize power consumption by using appropriate battery systems and efficient components, ensuring reasonable operating time between charges.
8. **Educational Value:** To create a project that serves as an excellent learning platform for understanding robotics, embedded systems, sensor integration, and automation principle

SYSTEM OVERVIEW

The robot integrates mechanical, electronic, and control components.

- Arduino Uno acts as the main controller.
- Two ultrasonic sensors detect obstacles and guide movement.
- RFID reader activates the robot when the authorized card is scanned.
- Motor driver (L298N) powers four DC motors that move the robot.
- Two vacuum fans collect dust during movement.
- Water pump sprays water and a wiper mops the floor.
- 12V battery powers high-load components, while a secondary battery runs the electronics.

HARDWARE COMPONENTS

1. **Arduino Uno:** The Arduino Uno serves as the central processing unit and controller of the entire system. It processes inputs from the ultrasonic sensors and RFID reader, executes the programmed logic for navigation and obstacle avoidance, and controls the motor driver for robot movement. Its 14 digital I/O pins and 6 analog inputs provide sufficient connectivity for all components. The Arduino's ease of programming using the Arduino IDE makes it ideal for this project.

2. **Ultrasonic Sensors (HC-SR04) - 2 units:** These sensors enable the robot to "see" its environment by emitting ultrasonic sound waves and measuring the time taken for echoes to return. Operating at 40 kHz frequency, they provide distance measurements from 2cm to 400cm with excellent accuracy. Two sensors are strategically positioned to provide wider coverage and more reliable obstacle detection from multiple angles.

3. **RFID Reader Module + RFID Card:** The Radio Frequency Identification system consists of an RFID reader (typically RC522 module) and one or more RFID cards or tags. This provides secure, contactless authentication for starting the robot. The reader operates at 13.56 MHz and communicates with Arduino via SPI protocol. Multiple cards can be programmed for authorized users.

4. **L298N Motor Driver:** This dual H-bridge motor driver module controls the speed and direction of the four DC motors. It can handle up to 2A per channel and operates with voltages from 5V to 35V. The module includes built-in voltage regulators and provides both direction and speed (PWM) control for the motors, enabling smooth navigation and turns.

5. **DC Motors with Wheels - 4 sets:** Four 12V DC motors provide the robot's mobility. Each motor is connected to a wheel, creating a four-wheel drive system that offers better traction and maneuverability. The motors are controlled in pairs (left side and right side) to enable forward, backward, left turn, and right turn movements.

6. **12V Battery:** A rechargeable 12V lead-acid or lithium-ion battery serves as the main power source for high-current components like motors and vacuum fans. The capacity (typically 7Ah or higher) determines the operating duration of the robot.

7. **Vacuum Fans - 2 units:** Two 12V DC centrifugal fans create suction for dust collection. These fans are positioned to cover maximum floor area during robot movement. They draw dust and small debris into a collection container through an intake system.

8. **Water Pump with Remote Control:** A 12V DC submersible water pump draws water from the onboard tank and sprays it ahead of the wiper. The wireless remote control (typically 433MHz RF transmitter-receiver) allows users to turn the pump on/off from a distance, providing flexibility in water usage.

9. **Sprinkler Pipe & Wiper:** The sprinkler pipe distributes water evenly across the cleaning path. Made of plastic tubing with small holes, it ensures uniform water distribution. The wiper (typically microfiber cloth or sponge material) is attached at the rear to mop up the sprayed water and clean the floor surface.

10. **Additional Components:**

- Relay Module: Controls high-power devices like the water pump safely through Arduino's low-voltage signals
- Battery Holder: Secures batteries in place and provides proper connections
- Main Switch: Enables complete system shutdown for safety and maintenance
- Jumper Wires: Connect various components according to the circuit design
- Chassis: The structural body that holds all components together, typically made from acrylic, wood, or lightweight metal

DESIGN AND DEVELOPMENT

The working of the device is divided into two different units.

A) Obstacle detection and Avoidance Unit

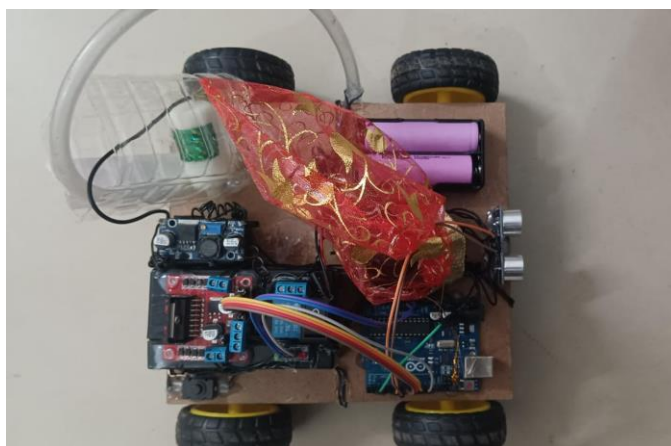
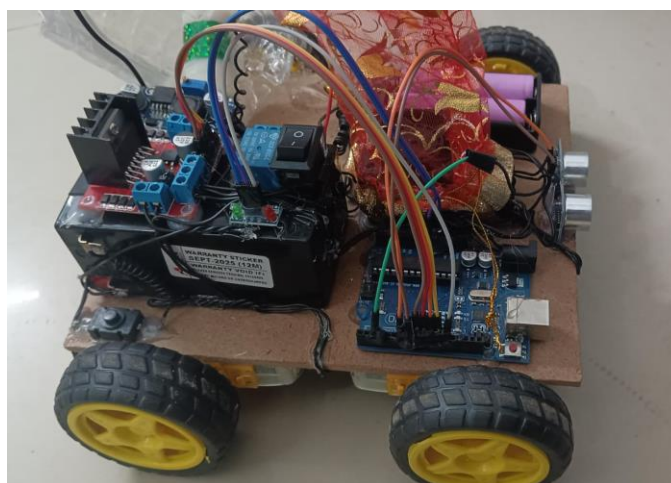
The ultrasonic sensor is used to detect an obstacle in its path. The ultrasonic sensor transmits ultrasonic waves which detect the presence of an obstacle in its given range. The robot sense or measures the distance of the obstacle with the help of an ultrasonic sensor and changes its direction accordingly. The sensing distance range of the robot is set by Arduino Uno programming and within this range, the robot senses the obstacle and turns back.

B) Cleaning Unit

Once the obstacle is successfully avoided by the robot, it starts its cleaning operation. If there is no obstacle in the path, the robot continues its cleaning. The cleaning unit includes two types of cleaning .i.e.

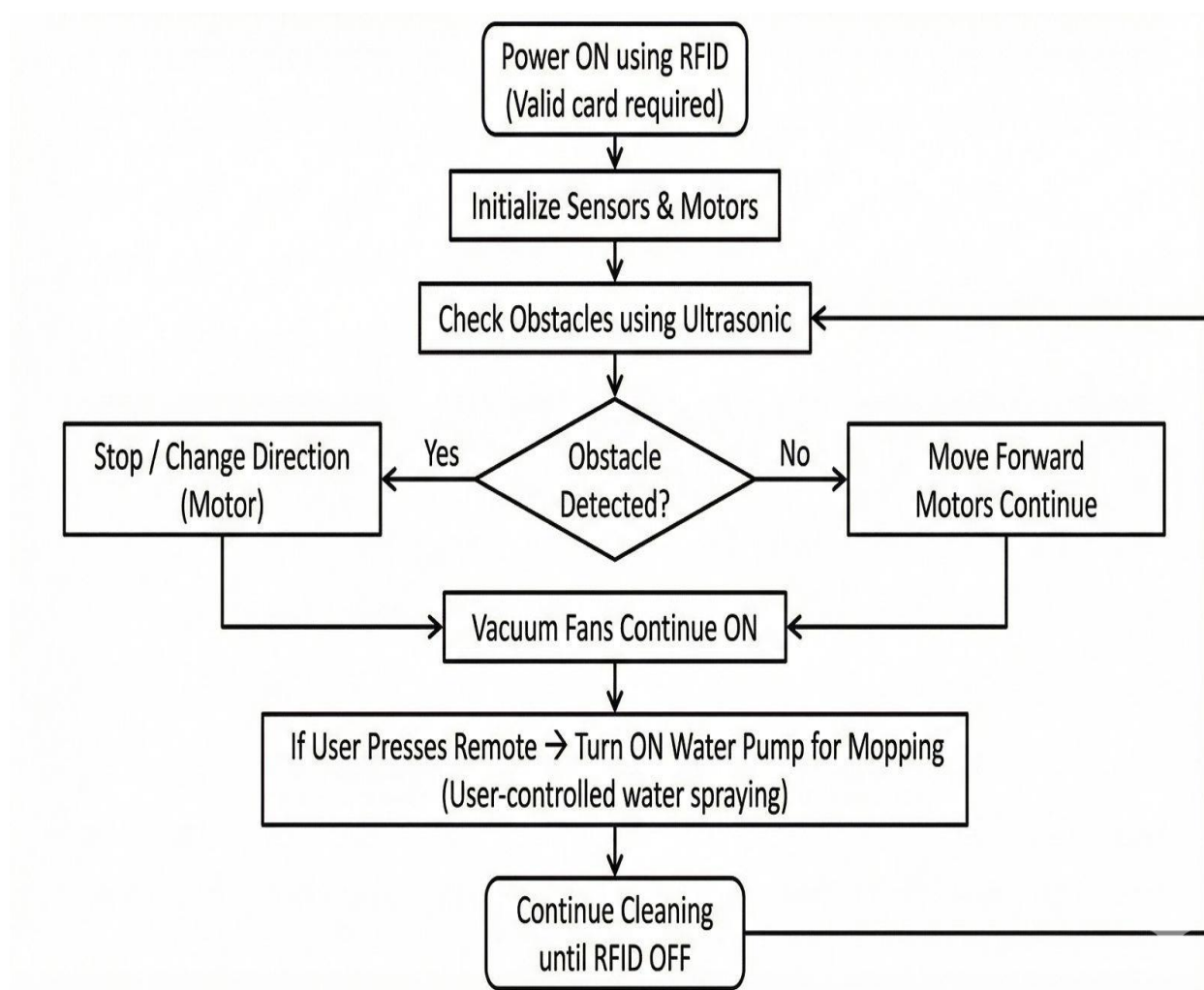
1)Wet Mopping: Stain cleaning is done by using a wet mopping. The mop/wipper is fitted at the back end of the robot. We used a 12V DC water pump that is deep into the water tank, so when the relay gets start we get water through the DC water pump for cleaning.

2)Vacuum Cleaning: Dust is clean by using a vacuum. By using a Vacuum cleaner dust can be easily sucked from the floor.



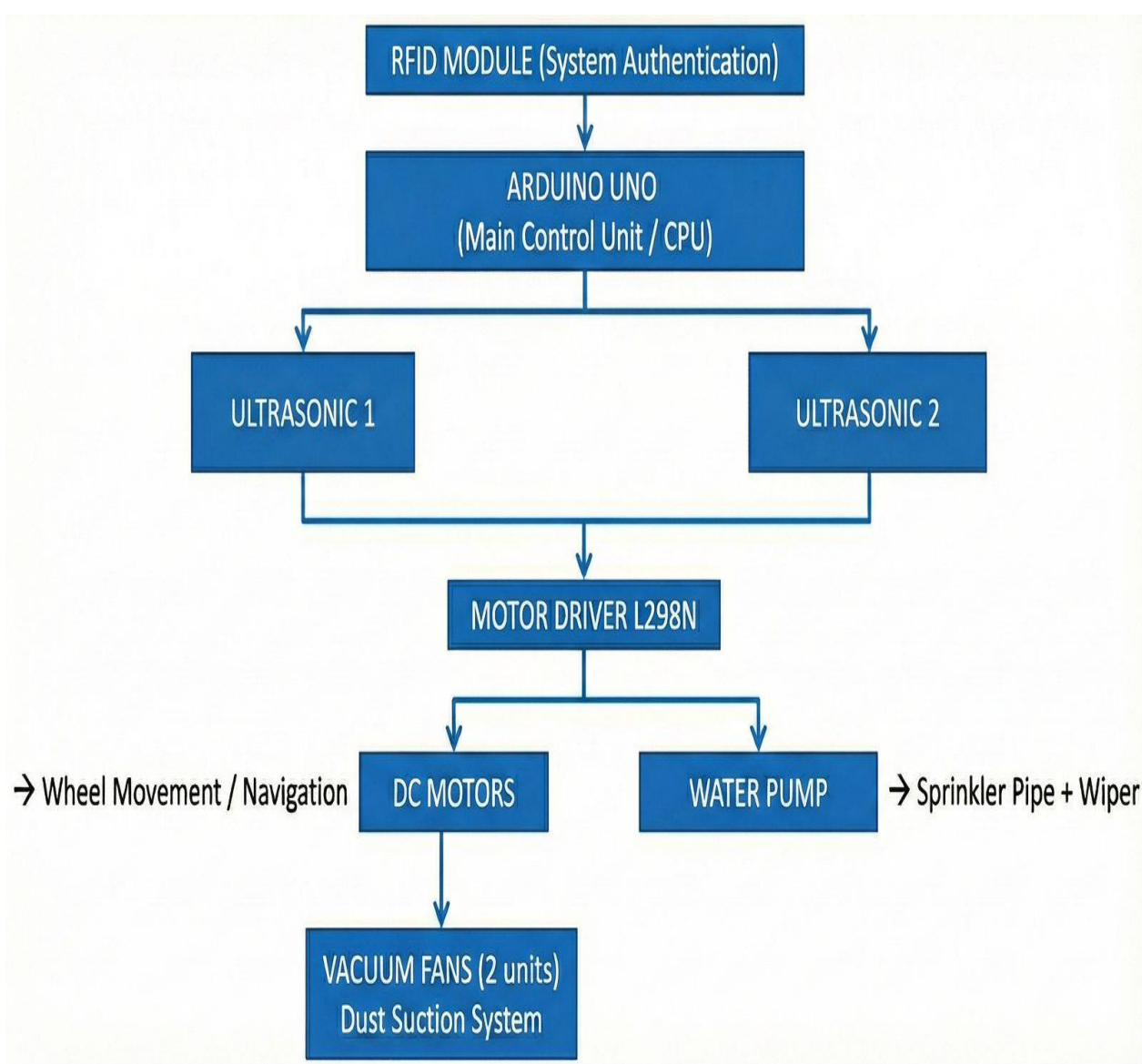
WORKING PRINCIPLE

After powering the robot using the RFID card, the ultrasonic sensors continuously detect obstacles. When an obstacle appears in front of the robot, Arduino changes the motor directions to avoid collision. The sensors measure the distance by calculating the time taken for ultrasonic waves to return after hitting an object, and if the distance falls below the programmed threshold (typically 15-20 cm), the Arduino immediately triggers the avoidance maneuver. The vacuum fans operate continuously to remove dust, and the water pump can be turned ON manually through the remote switch. Sprayed water is wiped automatically by the wiper during robot movement. The system maintains a cleaning cycle where the robot moves forward in straight lines until an obstacle is detected, then executes turning maneuvers (either left, right, or reverse) to find a clear path, ensuring maximum floor coverage without manual intervention.



CIRCUIT & CONNECTION DETAILS

The Arduino Uno works as the main controller. The ultrasonic sensors and RFID reader are connected to the Arduino to enable obstacle detection and controlled startup. The ultrasonic sensors use digital pins for Trigger and Echo signals to send and receive ultrasonic pulses for distance measurement. The L298N motor driver is connected to the Arduino and controls the four DC motors for robot movement. The vacuum fans receive power directly from the 12V battery through the main switch, while the water pump is activated using a switch and the wireless remote system. The relay module receives control signals from the Arduino's digital pin and switches the high-current circuit for the water pump, ensuring safe isolation between the low-voltage control circuit and high-power water pump. The sprinkler pipe and wiper are connected to the chassis to complete the mopping function. The wiring ensures that Arduino handles control logic while high-power devices are powered separately for safety.



COMPARISON WITH EXISTING CLEANING ROBOTS

Feature	Commercial Robots	Proposed Robot
Cost	₹20,000 – ₹38,000	₹4,000
Startup	App / Button	RFID authentication
Navigation	LIDAR / Camera	Ultrasonic sensors
Cleaning	Vacuum + Mop	Vacuum + Mop
Water Control	Automatic	Manual remote control
Repair	Difficult & costly	Easy & inexpensive

Comparison with Fully Automatic Mopping

Aspect	Fully Automatic Mopping (Commercial Robots)	Remote-Controlled Mopping (Proposed Robot)
User Control	None (mopping always ON once started)	Full control – user turns pump ON/OFF using remote
Floor Type Adaptation	Pre-programmed; may mop on unsuitable surfaces	User decides where mopping is required
Water Conservation	May over-mop / waste water	Highly efficient – water sprayed only when needed
Carpet Protection	Requires virtual mapping/No-mop zones	User prevents water on carpet manually
Battery Life	Pump always ON consumes more power	Better battery backup because pump runs only when needed
Complexity	Requires apps/sensors/software	Simple mechanism – no software dependency
Cost	Adds ₹5,000–₹9,000 to robot price	Almost free – included in ₹4,000 total build cost

Our proposed robot provides a cheaper and more easily repairable alternative, with core cleaning functionality at a significantly lower price.

APPLICATIONS

- **Household Indoor Cleaning:** The robot is ideal for daily cleaning of living rooms, bedrooms, kitchens, and hallways in residential homes. It can operate independently while family members are away at work or school, ensuring a clean home environment without manual effort. The dual vacuum and mopping functionality makes it suitable for maintaining hygiene in homes with children, pets, or elderly residents who are more susceptible to dust allergies and floor contaminants.
- **Classrooms & Laboratory Cleaning:** Educational institutions require frequent cleaning to maintain hygienic learning environments. This robot can clean classrooms after school hours, computer labs, and science laboratories where dust accumulation can damage sensitive equipment. The RFID-controlled access ensures only authorized staff can operate it, preventing student misuse.
- **Office Floor Maintenance:** Corporate offices, conference rooms, and cubicle areas can be cleaned efficiently during non-working hours. The robot's quiet operation and autonomous navigation make it perfect for maintaining professional workspace cleanliness without disturbing employees during working hours.
- **Hospital and Healthcare Facilities:** In healthcare settings where hygiene is critical, the robot can assist in cleaning waiting areas, corridors, and administrative sections. While not suitable for sterile operation rooms, it reduces the cleaning burden on hospital staff for general floor maintenance areas.
- **Elderly Assistance Systems:** For senior citizens or physically challenged individuals, bending and kneeling for floor cleaning can be painful or impossible. This robot provides an affordable assistive technology solution, promoting independent living and reducing dependency on caregivers for routine cleaning tasks.
- **Hotels and Guest Houses:** Hospitality industry can use these robots for cleaning guest room floors, lobbies, and corridor areas, improving cleaning efficiency and reducing labor costs while maintaining high cleanliness standards expected by guests.
- **Libraries and Reading Rooms:** These quiet spaces benefit from the robot's relatively silent operation, allowing floor cleaning without disturbing readers and researchers during operational hours.
- **Educational Robotics and Embedded Systems Learning:** Beyond practical cleaning applications, this project serves as an excellent teaching tool for engineering students to understand robotics fundamentals, sensor integration, microcontroller programming, motor control, and system integration principles in a hands-on, practical manner.

ADVANTAGES

- **Cost-Effective Smart Cleaning Solution:** At approximately ₹4,000, this robot costs only 10-20% of commercial alternatives (₹20,000-₹38,000), making advanced cleaning automation accessible to average households, small businesses, and educational institutions with limited budgets. The affordability democratizes access to robotic technology.
- **Dual Cleaning Functionality:** Unlike single-function cleaners, this robot performs both vacuum cleaning for dust removal and wet mopping for stain removal simultaneously. This dual functionality saves time, as users don't need to run separate devices or perform multiple cleaning passes, resulting in faster and more thorough cleaning.
- **Intelligent Automatic Obstacle Avoidance:** The ultrasonic sensor-based navigation system enables the robot to detect and avoid obstacles autonomously without human intervention. This prevents collisions with furniture, walls, and other objects, protecting both the robot and household items from damage while ensuring continuous cleaning operation.
- **RFID-Controlled Startup Prevents Unauthorized Use:** The authentication system ensures only authorized personnel can operate the robot, which is particularly valuable in institutional settings like schools, offices, and commercial spaces. This prevents theft, misuse by untrained individuals, and potential damage from improper operation.
- **User-Controlled Water Management:** The remote-controlled water pump gives users complete control over when and where mopping occurs. This prevents water damage to carpets, wooden furniture bases, or electronic equipment on the floor, while also promoting water conservation by eliminating unnecessary water usage.
- **Easy to Repair and Modify:** Built using standard, readily available components from local electronics markets, the robot can be easily repaired without expensive manufacturer service centers. Users with basic technical knowledge can replace damaged parts, upgrade components, or modify the design to suit specific needs, significantly reducing long-term maintenance costs.
- **Environmentally Friendly:** Reduces the need for disposable cleaning products and wipes, promoting sustainable cleaning practices. The rechargeable battery system eliminates the need for continuous power consumption during operation.
- **Customizable and Scalable:** The open-source Arduino platform and modular design allow users to customize the robot's behavior, add new features, or integrate additional sensors and functionalities according to specific requirements without being locked into proprietary systems.
- **Time-Saving Automation:** Frees up valuable time for users to focus on more productive activities rather than spending hours on repetitive cleaning tasks, improving overall quality of life and work-life balance.

LIMITATIONS

Not Suitable for Uneven or Outdoor Flooring: The robot is designed specifically for smooth, flat indoor surfaces like tiles, marble, hardwood, and linoleum. It cannot navigate stairs, steep ramps, high-pile carpets, or outdoor terrain with gravel, grass, or uneven pavement. The wheels may get stuck on thick rugs or door thresholds higher than 1-2 cm.

- **No Automatic Room Mapping or Navigation:** Unlike advanced commercial robots with LIDAR or camera-based mapping, this robot follows a random navigation pattern based on obstacle detection. It cannot create floor plans, remember cleaned areas, or optimize cleaning paths, which may result in some areas being cleaned multiple times while others might be missed.
- **Limited Battery Backup:** The operating duration depends on battery capacity and usage intensity. Typically, the robot can operate for 45-90 minutes on a single charge, which may not be sufficient for cleaning very large spaces. Unlike premium models with automatic docking stations, this robot requires manual recharging.
- **Basic Obstacle Detection:** The ultrasonic sensors can detect solid obstacles but may not recognize transparent glass, very thin objects like chair legs, or low-lying obstacles below sensor level. This can occasionally result in collisions or getting stuck under low-clearance furniture.
- **Manual Dust Container Emptying:** Unlike self-emptying commercial robots, this design requires users to manually remove and empty the dust collection container after cleaning sessions, adding a maintenance step to the process.
- **No Cliff Detection:** The current design lacks sensors to detect stairs or sudden drops in floor level, requiring users to block off stairways or elevated areas to prevent the robot from falling and getting damaged.
- **Water Tank Capacity Limitations:** The onboard water tank has limited capacity and requires refilling for extended cleaning sessions or large areas, interrupting the autonomous cleaning process.
- **Noise Level:** While quieter than traditional vacuum cleaners, the dual vacuum fans do generate noticeable noise (approximately 60-70 dB), which may be disturbing in noise-sensitive environments like bedrooms during sleep hours or libraries.
- **Limited Smart Features:** Lacks modern connectivity features like WiFi, mobile app control, scheduling, or voice assistant integration that are becoming standard in contemporary smart home devices.

FUTURE SCOPE

- **IoT and Mobile App-Based Robot Control:** Integrate ESP8266 or ESP32 WiFi modules to enable Internet connectivity, allowing users to control, monitor, and schedule cleaning sessions remotely through dedicated smartphone applications. Users could start cleaning while away from home, receive notifications when cleaning is complete, and check battery status in real-time.
- **Self-Charging Docking Station:** Develop an automatic docking station that the robot can autonomously locate and return to when battery levels are low. This eliminates manual recharging and enables truly autonomous operation where the robot can clean multiple times throughout the day without human intervention.
- **Voice Assistant Integration:** Incorporate compatibility with popular voice assistants like Amazon Alexa, Google Assistant, or Apple Siri, allowing users to control the robot through voice commands such as “Start cleaning the living room” or “Pause the robot.”
- **Artificial Intelligence and Machine Learning:** Implement AI algorithms that learn user preferences, cleaning patterns, and room layouts over time. The robot could automatically adjust cleaning intensity based on floor type, recognize frequently dirty areas, and optimize cleaning schedules based on household routines.
- **Enhanced Suction Power and Larger Dirt Container:** Upgrade to more powerful vacuum motors with higher suction capacity (measured in Pascals) to handle heavier debris, pet hair, and deep-cleaning requirements. Increase the dust container volume to reduce emptying frequency.
- **Cliff and Drop Detection Sensors:** Add infrared or ultrasonic sensors pointed downward to detect stairs, ledges, and sudden floor level changes, automatically reversing direction to prevent falls and damage.
- **Automatic Water Tank Refilling System:** Design a docking station with water reservoir that automatically refills the robot’s water tank when it docks, enabling continuous mopping operation for larger areas.
- **UV-C Sterilization Feature:** Integrate ultraviolet-C LED lights underneath the robot to sterilize floors by killing bacteria, viruses, and other microorganisms during cleaning, particularly valuable for healthcare facilities and homes with young children.
- **Solar Panel Integration:** Add small solar panels on the robot’s top surface to supplement battery charging, extending operating duration and improving energy efficiency.
- **Edge and Corner Cleaning Mechanism:** Add side brushes or specialized corner-cleaning attachments to improve cleaning effectiveness along walls, corners, and edges where the circular/rectangular robot body cannot reach.

CONCLUSION

The autonomous vacuum and mopping robot successfully demonstrates that advanced cleaning automation can be achieved at a fraction of the cost of commercial alternatives without compromising on essential functionality. This project validates the feasibility of developing affordable, practical robotic solutions that genuinely improve quality of life for average households and small institutions.

Through systematic integration of Arduino Uno microcontroller, ultrasonic sensors for obstacle detection, RFID-based security authentication, dual vacuum fans, and remote-controlled mopping system, the robot delivers comprehensive floor cleaning capabilities at an estimated cost of just ₹4,000. This represents approximately 80-90% cost savings compared to commercial robotic cleaners priced between ₹20,000 and ₹38,000, making the technology accessible to a much broader user base.

The project successfully addresses multiple real-world challenges: reducing the physical burden of manual cleaning, saving valuable time, providing cleaning solutions for elderly and physically challenged individuals, and offering a repairable alternative to expensive proprietary systems. The RFID authentication prevents unauthorized use in institutional settings, while the remote-controlled water pump provides intelligent water management that prevents damage to sensitive surfaces and conserves resources.

Beyond its immediate practical utility, this robot serves as an excellent educational platform for understanding fundamental concepts in robotics, embedded systems, sensor integration, motor control, and automation engineering. The design philosophy emphasizes accessibility and maintainability by utilizing standard, readily available components from local markets, allowing users with basic technical knowledge to perform repairs and customizations without expensive manufacturer servicing.

Testing and operation of the prototype have demonstrated effective cleaning performance on various indoor floor surfaces including tiles, marble, and hardwood. The obstacle avoidance system works reliably in typical household environments, and the dual cleaning functionality provides thorough cleaning in a single pass.

While the current design has limitations such as lack of advanced mapping and limited battery duration, these represent opportunities for future enhancement. The integration of IoT connectivity, mobile app control, advanced LIDAR-based mapping, and artificial intelligence can transform this prototype into a commercially viable product competitive with established brands.

In conclusion, this autonomous vacuum and mopping robot represents a successful synthesis of affordable hardware, intelligent programming, and practical engineering design. It proves that innovation doesn't require expensive resources—creativity, systematic problem-solving, and effective utilization of available technology can produce solutions that genuinely benefit society. With further development and refinement, this prototype has strong potential to evolve into a marketable product that brings the benefits of cleaning automation to millions of households and institutions that currently cannot afford commercial alternatives.

REFERENCES

- Manya Jain, Pankaj Singh Rawat “Automatic Floor Cleaner” International research journal of Engineering and Technology (IRJET) Volume:04 Issue:04 | Apr -2017 eISSN: 2395 -0056 p-ISSN: [2395-0072](#) (tel:[2395-0072](#))
- Yong-Joo Oh and Y. Watanabe "Development of the small robot for home floor cleaning", Proceedings of the 41st SICE Annual Conference, 5-7 Aug, SICE 2002